Project Communication & Quality

- Concluding topics

Topic # 13

- Chapter 8, 10 – Schwalbe

Communications Management Plan Contents

1. Stakeholder communications requirements
2. Information to be communicated, including format, content, and level of detail
3. Who will receive the information and who will produce it
4. Suggested methods or technologies for conveying the information
5. Frequency of communication
6. Escalation procedures for resolving issues
7. Revision procedures for updating the communications management plan
8. A glossary of common terminology

Sample Stakeholder Analysis for Project Communications

Stakeholders	Document Name	Document Format	Contact Person	Due
Customer management	Monthly status report	Hard copy and meeting	Tina Erndt, Tom Silva	First of month
Customer business staff	Monthly status report	Hard copy	Julie Grant, Sergey Cristobal	First of month
Customer technical staff	Monthly status report	E-mail	Li Chau, Nancy Michaels	First of month
Internal management	Monthly status report	Hard copy and meeting	Bob Thomson	First of month
Internal business and technical staff	Monthly status report	Intranet	Angie Liu	First of month
Training subcontractor	Training plan	Hard copy	Jonathan Kraus	November 1
Software subcontractor	Software implementation plan	E-mail	Najwa Gates	June 1

Comments: Put the titles and dates of documents in e-mail headings and have recipients acknowledge receipt.

Classifications for Communication Methods

- *Interactive communication*: Two or more people interact to exchange information via meetings, phone calls, or video conferencing. Most effective way to ensure common understanding
- *Push communication*: Information is sent or pushed to recipients without their request via reports, e-mails, faxes, voice mails, and other means. Ensures that the information is distributed, but does not ensure that it was received or understood
- *Pull communication*: Information is sent to recipients at their request via Web sites, bulletin boards, e-learning, knowledge repositories like blogs, and other means

Media Choice Table

KEY: 1 = EXCELLENT 2 = ADEQUATE 3 = INAPPROPRIATE

How well medium is suited to:	Hard Copy	Telephone Call	Voice Mail	E-mail	Meeting	Web Site
Assessing commitment	3	2	3	3	1	3
Building consensus	3	2	3	3	1	3
Mediating a conflict	3	2	3	3	1	3
Resolving a misunderstanding	3	1	3	3	2	3
Addressing negative behavior	3	2	3	2	1	3
Expressing support/appreciation	1	2	2	1	2	3
Encouraging creative thinking	2	3	3	1	3	3
Making an ironic statement	3	2	2	3	1	3
Conveying a reference document	1	3	3	3	3	1
Reinforcing one s authority	1	2	3	3	1	2
Providing a permanent record	1	3	3	1	3	1
Maintaining confidentiality	2	1	2	3	1	3
Conveying simple information	3	2	1	1	2	3
Asking an informational question	3	2	1	1	3	3
Making a simple request	3	3	1	1	3	3
Giving complex instructions	3	3	3	2	1	2
Addressing many people	2	3	3 or 1*	2	3	1

Galati, Tess. Email Composition and Communication (EmC2) Practical Communications, Inc. (www.praccom.com) (2001).
*Depends on system functionality

Reporting Performance

Performance reporting keeps stakeholders informed about how resources are being used to achieve project objectives

- **Status reports** describe where the project stands at a specific point in time
- **Progress reports** describe what the project team has accomplished during a certain period of time
- **Forecasts** predict future project status and progress based on past information and trends

Running Effective Meetings

- Determine if a meeting can be avoided
- Define the purpose and intended outcome of the meeting
- Determine who should attend the meeting
- Provide an agenda to participants before the meeting
- Prepare handouts and visual aids, and make logistical arrangements ahead of time
- Run the meeting professionally
- Set the ground rules for the meeting
- Build relationships

Using E-Mail, Instant Messaging, Texting, and Collaborative Tools Effectively

- Make sure that e-mail, instant messaging, texting, or collaborative tools are an appropriate medium for what you want to communicate
- Be sure to send information to the right people
- Use meaningful subject lines and limit the content of emails to one main subject, and be as clear and concise as possible
- Be sure to authorize the right people to share and edit your collaborative documents

Sample Collaborative Tools

- A **SharePoint portal** allows users to create custom Web sites to access documents and applications stored on shared devices
- **Google Docs** allow users to create, share, and edit documents, spreadsheets, and presentations online
- A **wiki** is a Web site designed to enable anyone who accesses it to contribute or modify Web page content

Sample Template for a Project Description

Project X Descripton

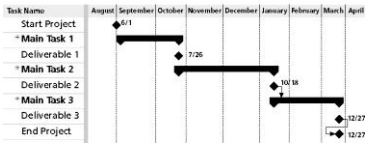
Objective: Describe the objective of the project in one or two sentences. Focus on the business benefits of doing the project.

Scope: Briefly describe the scope of the project. What business functions are involved, and what are the main products the project will produce?

Assumptions: Summarize the most critical assumptions for the project.

Cost: Provide the total estimated cost of the project. If desired, list the total cost each year.

Schedule: Provide summary information from the project's Gantt chart, as shown. Focus on summary tasks and milestones.



Sample Template for a Monthly Progress Report

I. Accomplishments for Month of January (or appropriate month):
- Describe most important accomplishments. Relate to project's Gantt chart.
- Describe other important accomplishments, one bullet for each. If any issues were resolved from the previous month, list them as accomplishments.

II. Plans for February (or following month):
- Describe most important items to be accomplished in the next month. Again, relate tothe project's Gantt chart.
- Describe other important items to accomplish, one bullet for each.

III. Issues: Briefly list important issues that surfaced or are still important. Managers hate surprises and want to help the project succeed, so be sure to list issues.

IV. Project Changes (Date and Description): List any approved or requested changes to the project. Include the date of the change and a brief description.

Table 10-4. Final Project Documentation Items

I.	Project description
II.	Project proposal and backup data (request for proposal, statement of work, proposal correspondence, and so on)
III.	Original and revised contract information and client acceptance documents
IV.	Original and revised project plans and schedules (WBS, Gantt charts and network diagrams, cost estimates, communications management plan, etc.)
V.	Design documents
VI.	Final project report
VII.	Deliverables, as appropriate
VIII.	Audit reports
IX.	Lessons-learned reports
X.	Copies of all status reports, meeting minutes, change notices, and other written and electronic communications

Lessons Learned Reports

- The project manager and project team members should each prepare a **lessons-learned report**
  - A reflective statement that documents important things an individual learned from working on the project
- The project manager often combines information from all of the lessons-learned reports into a project summary report

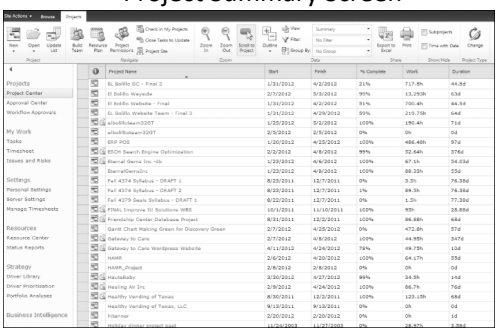
Project Archives

- It is also important to organize and prepare project archives
- **Project archives** are a complete set of organized project records that provide an accurate history of the project
- These archives can provide valuable information for future projects as well

Project Web Sites

- Many project teams create a project Web site to store important product documents and other information
- Can create the site using various types of software, such as enterprise project management software

Microsoft's Project Web Application Master Project Summary Screen



What Went Wrong?

- In 1986, two hospital patients died after receiving fatal doses of radiation from a Therac 25 machine after a software problem caused the machine to ignore calibration data
- Britain's Coast Guard was unable to use its computers for several hours in May 2004 after being hit by the Sasser virus, which knocked out the electronic mapping systems, e-mail, and other computer functions, forcing workers to revert to pen, paper, and radios
- More than 100 incidents of lost or stolen financial information were reported over the past year, including personal information of 1.2 federal employees, 200,000 online trading customers, and 33,000 Air Force officers

What Is Project Quality?

- The International Organization for Standardization (ISO) defines **quality** as "the degree to which a set of inherent characteristics fulfills requirements" (ISO9000:2000)
- Other experts define quality based on:
  - **Conformance to requirements**: the project's processes and products meet written specifications
  - **Fitness for use**: a product can be used as it was intended

Defining Quality – Quality Framework by ISO-9126(ISO, 2001)

- Various so called "ilities" …. The Quality Characteristics

1. Functionality: A set of attributes that a set of tasks & their specified properties. These functions or tasks satisfy stated or implied needs.
2. Reliability: A set of attributes that bear on the capability of the s/w to maintain its level of performance under stated conditions for a specified period of time.
3. Usability: A set of attributes that bear on effort needed for use and on the individual assessment of such use.
4. Efficiency: A set of attributes that bear on the relationship b/w the level of performance of the s/w & the amount of resources used under specified conditions
5. Maintainability: The set of attributes that bear on the effort needed to make specified modifications.
6. Portability: A set of attributes that bear on the ability of s/w to be transferred from one environment to another.

What Is Project Quality Management?

- **Project quality management** ensures that the project will satisfy the needs for which it was undertaken
- Processes include:
  - **Quality planning**: identifying which quality standards are relevant to the project and how to satisfy them
  - **Quality assurance**: periodically evaluating overall project performance to ensure the project will satisfy the relevant quality standards
  - **Quality control**: monitoring specific project results to ensure that they comply with the relevant quality standards

Scope Aspects of IT Projects

- It is often difficult for customers to explain exactly what they want in an IT project. Important scope aspects of IT projects that affect quality include:
  - **Functionality** is the degree to which a system performs its intended function
    - **Features** are the system's special characteristics that appeal to users. It is important to specify which are required and which are optional
  - **System outputs** are the screens and reports the system generates. Need to define clearly what they look like
  - **Performance** addresses how well a product or service performs the customer's intended use.
    - Need to know volumes of data and transactions, number of simultaneous users, required response time, etc.
  - **Reliability** is the ability of a product or service to perform as expected under normal conditions (customers must define expected level of service)
    - **Maintainability** addresses the ease of performing maintenance on a product

Quality Assurance

- **Quality assurance** includes all the activities related to satisfying the relevant quality standards for a project
  - Another goal of quality assurance is continuous quality improvement
- **Benchmarking** generates ideas for quality improvements by comparing specific project practices or product characteristics to those of other projects or products within or outside the performing organization
- A **quality audit** is a structured review of specific quality management activities that help identify lessons learned that could improve performance on current or future projects
  - Perfomed by in-house auditors or third parties

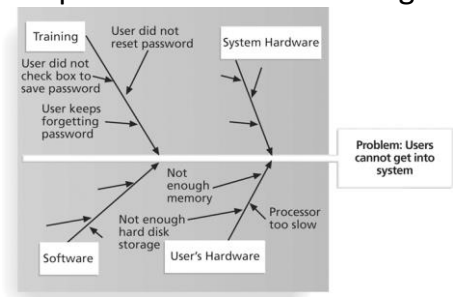
Quality Control

- Although one of the main goals of QC is to improve quality, its main outcomes are:
  - ***Acceptance decisions***- are the products/services acceptable or should they be rejected and rework is then necessary
  - ***Rework*** – action taken to bring rejected items into compliance with products specs. Can be very expensive
  - ***Process adjustments*** – correct or prevent further quality problems based on quality control measurements (purchase faster server if response time is too slow)

Tools & Techniques for Quality Control

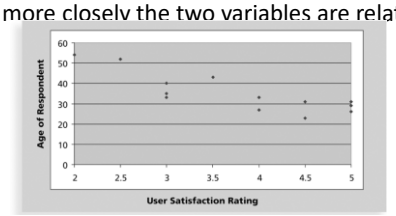
- **Cause-and-effect diagrams** trace complaints about quality problems back to the responsible production operations
  - They help you find the root cause of a problem
  - Also known as **fishbone** or **Ishikawa diagrams**
  - Can also use the **5 whys** technique where you repeat the question "Why" (five is a good rule of thumb) to peel away the layers of symptoms that can lead to the root cause
    1. Why the users can not get into the system
    2. Why they keep forgetting passwords
    3. Why didn't they reset their passwords
    4. Why didn't they check the box to save their password, etc.

Sample Cause-and-Effect Diagram



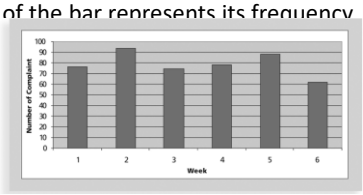
Scatter Diagram

- A **scatter diagram** helps to show if there is a relationship between two variables
- The closer data points are to a diagonal line, the more closely the two variables are related



Histograms

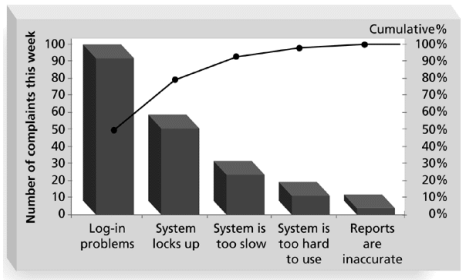
- A **histogram** is a bar graph of a distribution of variables
- Each bar represents an attribute or characteristic of a problem or situation, and the height of the bar represents its frequency



Pareto Charts

- A **Pareto chart** is a histogram that can help you identify and prioritize problem areas
  - The variables are ordered by frequency of occurrence to help identify the key contributors that account for most quality problems (hopefully following the 80-20 rule)
- **Pareto analysis** is also called the 80-20 rule, meaning that 80 percent of problems are often due to 20 percent of the causes
- In the following chart, Log-in Problems account for about 55% of the complaints and together with System lock-ups accounts for about 80%
  - Fixing these two problems can greatly reduce the volume of compalints
  - Small problems should be investigated before addressing them in case the user is in error

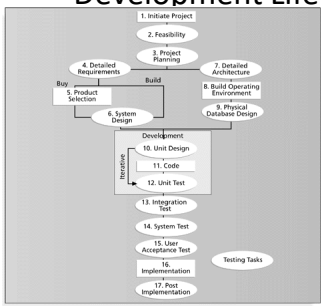
Sample Pareto Diagram



Testing

- Many IT professionals think of testing as a stage that comes near the end of IT product development
- Testing should be done during almost every phase of the IT product development life cycle

Testing Tasks in the Software Development Life Cycle



- **One way of portraying the systems life cycle**
- **Shows 17 main tasks involved in a s/w development project and shows their realtionship to each other**

Types of Tests

- **Unit testing** tests each individual component (often a program) to ensure it is as defect-free as possible
- **Integration testing** occurs between unit and system testing to test functionally grouped components
- **System testing** tests the entire system as one entity
- **Acceptance Testing**
- **User acceptance testing** is an independent test performed by end users prior to accepting the delivered system

The Cost of Quality

- The **cost of quality** is the cost of conformance plus the cost of nonconformance
  - **Conformance** means delivering products that meet requirements and fitness for use
  - **Cost of nonconformance** means taking responsibility for failures or not meeting quality expectation

Five Cost Categories Related to Quality

- **Prevention cost**: cost of planning and executing a project so it is error-free or within an acceptable error range
- **Appraisal cost**: cost of evaluating processes and their outputs to ensure that a project is either error-free or within an acceptable error range
- **Internal failure cost**: cost incurred to correct an identified defect before the customer receives the product (rework, inventory costs due to defects, premature failure of products)
- **External failure cost**: cost that relates to all errors not detected and corrected before delivery to the customer (warranty costs, product liability suits, future business losses)
- **Measurement and test equipment costs**: capital cost of equipment used to perform prevention and appraisal activities

END OF TOPIC 13

-COMING UP!!!!!!
- Project Stakeholder & Procurement Management
-Concluding topic
-Semester Exam
-Project Submission & Demo

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